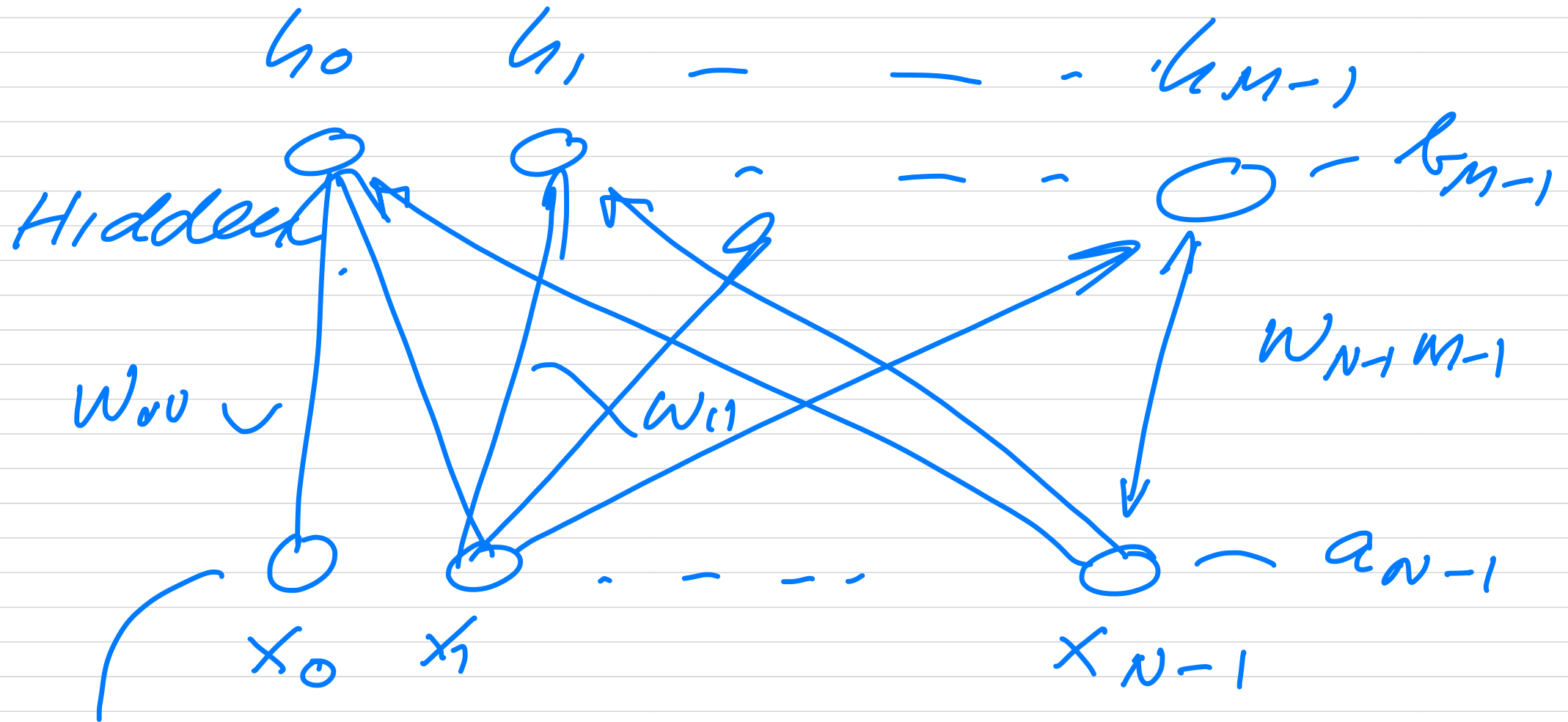


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visible

Binary-Binary RBM

$$x_i = \{0, 1\}_{2^N}$$

$$h_j = \{0, 1\}_{2^M}$$

$$E(x, h; \theta) = - \sum_{n=0}^{N-1} a_n' x_n' - \sum_{j=0}^{M-1} b_j' h_j' - \sum_{i,j} x_n' w_{ij}' h_j'$$

$$= - (a^T x + b^T h + x^T W h)$$

$$p(x, h; \theta) = \frac{e^{-E(x, h; \theta)}}{Z(\theta)}$$

$$Z(\theta) = \sum_{\underbrace{x, h}_{\text{all possible config}}} e^{-E(x, h; \theta)}$$

$$N=2 \quad \wedge \quad M=2$$

$$X = \{0,0\}, \{0,1\}, \{1,0\}, \{1,1\}$$

$$P(X; \theta) = \prod_{x_i \in D_-} P(x_i; \theta)$$

$$\hat{\theta} = \underset{\theta \in \mathbb{R}^P}{\operatorname{argmax}} \log P(X; \theta)$$

$$\theta = \{\vec{a}, \vec{b}, w\}$$

optimization

$$\mathbb{E} \log p(x; \theta) = 0$$

Standard RBM approach

However, we will use a
NGS-RBM

$$|y_i|^2 = p(x; \theta)$$

$$\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^P} \mathbb{E}[E_L(\theta)]$$

Conditional probability

$$P(h|x) \rightarrow$$

$$P(h_j = 1|x) \wedge P(h_j = 0|x)$$

$$P(h|x) \geq 0.5 \text{ then } h_j = 1 \\ \text{else } h_j = 0$$

$$P(h_j|x) \propto \text{sigmoid}$$

HO in 2 dim

— Bosons $N = 2$

$$\begin{array}{l} \text{—————} \quad n_x = 1, n_y = 0 \quad \wedge \quad n_x = 0, n_y = 1 \\ \text{— } \bigcirc \bigcirc \text{ —} \quad n_x n_y = 0 \end{array}$$

$$E_{N=2} = 2 \cdot \hbar \omega (n_x + n_y + 1)$$

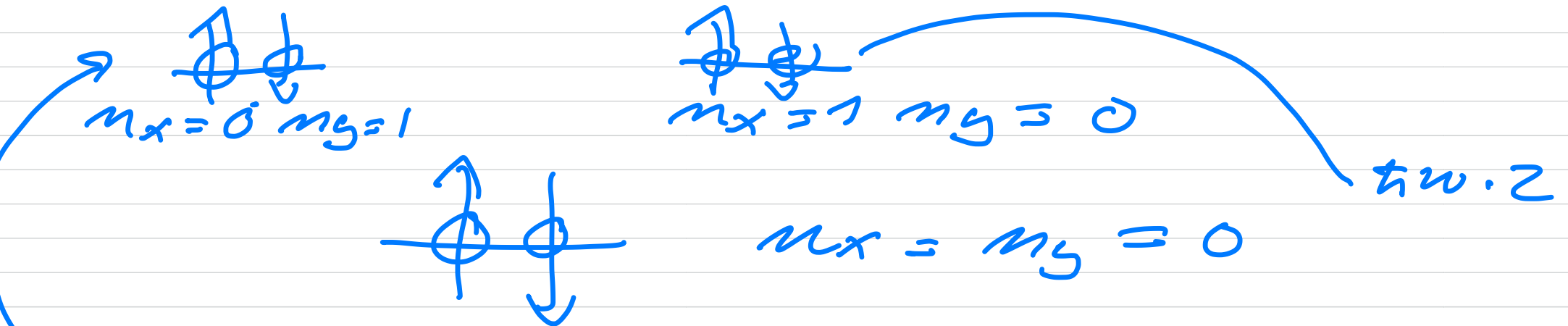
$$E_N = N \hbar \omega (n_x + n_y + 1)$$

$$n_x = n_y = 0$$

$$E_0 = N \cdot \hbar \omega$$

HO in 2 Dim

Fermions $N=2$ (spin $\frac{1}{2}$)



$N=6$

Bosons

$$E_0 = 6 \cdot \hbar \omega$$

$$\begin{aligned} \text{Fermions } E_0 &= 2 \cdot \hbar \omega + 8 \hbar \omega \\ &= 10 \hbar \omega \end{aligned}$$

$N = 12$ for fermions

3 tw

$n_x = 0, n_y = 2$
 $\uparrow \downarrow$

$n_x = n_y = 1$
 $\uparrow \downarrow$

$n_x = 2, n_y = 0$
 $\uparrow \downarrow$

2 tw

$n_x = 0, n_y = 1$
 $\uparrow \downarrow$

$n_x = 1, n_y = 0$
 $\uparrow \downarrow$

1 tw

1 tw

$\uparrow \downarrow$

$n_x = n_y = 0$

